

Hermes Agent Architecture

Public-safe architecture snapshot generated at 2026-09-03T06:15:53-04:00.

Source of truth: local Hermes configuration and runtime status on the operator Linux host.

Sensitive values, private chat IDs, OAuth headers, personal finance/client workflow names, and tokens are intentionally omitted or grouped.

Executive Summary

Hermes Agent runs as a **Linux-primary, multi-platform AI operations hub**. The gateway is a systemd service that connects Telegram, Mattermost, WhatsApp, API Server, Home Assistant, MCP servers, cron jobs, skills, memory/context stores, and remote Windows operations.

The current architecture adds a **durable Kanban orchestration layer**: `default` remains the interactive/orchestrator profile, while `researcher`, `worker`, and `reviewer` are launched as ephemeral Kanban worker-runs for durable research, implementation, and verification tasks. Safe post-session automation generates local review artifacts (session summaries, skill-mining candidates, trace-derived code graphs) without silently installing skills or uploading traces.

The default model remains `openai-codex / gpt-5.5`. Local/experimental providers such as **LM Studio** and **Free Kimi** are configured as optional providers only; they are not the default route for Telegram or production workflows.

High-Level Architecture

[!Hermes Architecture](#)

Runtime Topology

Layer	Components	Notes
User channels	Telegram, Mattermost, WhatsApp, API Server	Gateway adapters route inbound messages into the same Hermes agent core.
Agent core	Hermes Agent gateway + CLI + tool loop	Runs on Linux under <code>hermes-gateway.service</code> .
Models	OpenAI Codex primary, Copilot fallback, optional LM Studio, optional Free Kimi proxy	Provider selection is explicit; experimental endpoints are not default.
Tools	Terminal, files, browser, web, vision, TTS, cron, delegation, Home Assistant, GBrain, NotebookLM, CodeGraph	Enabled toolsets differ by platform/session but the gateway exposes the main operational surface.
Knowledge	GBrain, NotebookLM, session search, imported Claude context, skills	Long-term memory and research context are split across structured stores and markdown skills.
Automation	Hermes cron scheduler, no-agent scripts, hooks, plugins, post-session automation	Backups, monitoring, GitHub publication, finance snapshots, smart-home logging, news/media digests, watchdogs, session summaries, skill-mining reviews, and trace graphs.
Durable orchestration	Kanban dispatcher + <code>default/researcher/worker/reviewer</code> profiles	The gateway dispatcher claims ready tasks and launches profile-specific worker-runs; <code>stopped</code> profiles are normal idle state.
Operational surfaces	Tasks, skills, rules, hooks, plugins, MCP servers, toolsets	Documented as public-safe counts/categories plus selected non-sensitive examples.
Remote systems	Windows workstation over SSH, Home Assistant, Google Drive/GitHub, local LM Studio	Linux remains the control plane; Windows is operated remotely when needed.

Operational Surface Inventory

Surface	Detected public-safe state	Notes
Scheduled tasks / cron	78 jobs; 39 no-agent script jobs; 0 agent-backed jobs	Exact private task names are grouped by category.
Skills	391 detected skill files across 28 categories	Private/client-sensitive skill names are omitted from examples.
Hooks / webhooks	shell allowlist present: False; allowlist entries: 0; plugin hook manifests: 0	Hook command bodies are not published.
Plugins	58 visible plugin rows captured; enabled estimate 5	Descriptions omitted to avoid leaking credential/env surfaces.
MCP servers	11 configured MCP servers	GBrain, NotebookLM, CodeGraph are the active core MCP surfaces.

Scheduled tasks / cron categories

Category	Active jobs	Public-safe purpose
Backup & sync	5	Protect configuration, repositories, databases, and knowledge stores.
GitHub & publishing	6	Maintain GitHub/publication surfaces and repo health digests.
Home automation	2	Log smart-home/home-environment telemetry.
Knowledge & memory	7	Keep GBrain/memory/context stores healthy and up to date.
Media/news monitoring	2	News, RSS, YouTube, and briefing pipelines.
Other scheduled automation	39	Other local automation jobs.
Private finance automation	5	Private finance workflow snapshots; details omitted from public docs.
Reliability watchdogs	12	Auto-healing, environment guards, timeout/watchdog checks.

Skills surface

Hermes currently has a broad skill surface. The public inventory lists category counts and non-sensitive examples only.

Skill category	Count
.archive	12
android	2
apple	9
autonomous-ai-agents	22
creative	35
data-science	1
devops	28
ecc-imports	4
email	5
external	7
gaming	2
github	9
mcp	2
media	8
mlops	18
note-taking	4
operations	3
personal	9
productivity	45
red-teaming	1
research	33
security	4
smart-home	4
social-media	2
software-development	100
uncategorized	19
web	2

Skill category	Count
web-development	1

Public-safe skill examples:

Skill	Category	Public-safe description
obsidian	note-taking	Read, search, create, and edit notes in the Obsidian vault.
comfyui	creative	Generate images, video, and audio with ComfyUI — install, launch, manage nodes/models, run workflows with parameter injection. Uses the offi
[REDACTED]	creative	Article illustrations: type × style × palette consistency.
humanizer	creative	Humanize text: strip AI-isms and add real voice.
pixel-art	creative	Pixel art w/ era palettes (NES, Game Boy, PICO-8).
ascii-video	creative	ASCII video: convert video/audio to colored ASCII MP4/GIF.
manim-video	creative	Manim CE animations: 3Blue1Brown math/algo videos.
architecture-diagram	creative	Dark-themed SVG architecture/cloud/infra diagrams as HTML.
ascii-media	creative	Create ASCII still art, banners, image conversions, and animated ASCII video/audio visuals from one umbrella workflow.
claude-design	creative	Design one-off HTML artifacts (landing, deck, prototype).
pretext	creative	Build creative browser demos with DOM-free text layout.
hyperframes	creative	Create HTML-based video compositions, animated title cards, social overlays, captioned talking-head videos, audio-reactive visuals, and shad
apple-design	creative	Design or review Apple-inspired web interactions with direct manipulation, interruptible spring motion, velocity handoff, momentum, rubber-b
[REDACTED]	creative	Songwriting craft and Suno AI music prompts.
p5js	creative	p5.js sketches: gen art, shaders, interactive, 3D.
reference-safe-design	creative	Use when a user supplies websites, screenshots, brand campaigns, moodboards, copy, motion, or assets and wants either an originality audit o
baoyu-comic	creative	Knowledge comics (知识漫画): educational, biography, tutorial.
touchdesigner-mcp	creative	Control TouchDesigner via twozero MCP.
ideation	creative	Generate project ideas via creative constraints.
design-md	creative	Author/validate/export Google
excalidraw	creative	Hand-drawn Excalidraw JSON diagrams (arch, flow, seq).
popular-web-designs	creative	54 real design systems (Stripe, Linear, Vercel) as HTML/CSS.

Skill	Category	Public-safe description
concept-diagrams	creative	Generate flat, minimal light/dark-aware SVG diagrams as standalone HTML files, using a unified educational visual language with 9 semantic c
blender-mcp	creative	Control Blender directly from Hermes via socket connection to the blender-mcp addon. Create 3D objects, materials, animations, and run arbit
baoyu-infographic	creative	Infographics: 21 layouts x 21 styles (信息图, 可视化).

Hooks, webhooks, and plugin hook manifests

Hermes has multiple hook-related surfaces: shell-hook allowlists, webhook subscriptions, and imported Claude/Claude plugin hook manifests. The public repo records only surface/count information, not hook command bodies.

Hook manifest surface
none detected

Plugin surface

Plugin	Status
browser-browser-use	not enabled
browser-browserbase	not enabled
browser-firecrawl	not enabled
chronos	not enabled
basic	not enabled
drain	not enabled
nous	not enabled
self-hosted	not enabled
disk-cleanup	not enabled
google_meet	not enabled
deepinfra	not enabled
fal	not enabled
krea	not enabled
openai	not enabled
openai-codex	not enabled
openrouter	not enabled
xai	not enabled
langfuse	not enabled
a2a-platform	not enabled
buzz-platform	not enabled
dingtalk-platform	not enabled
discord-platform	not enabled
email-platform	not enabled
feishu-platform	not enabled
google_chat-platform	not enabled

Plugin	Status
irc-platform	not enabled
line-platform	not enabled
matrix-platform	not enabled
mattermost-platform	not enabled
ntfy-platform	not enabled

Low-Level Surface Files

The repository includes dedicated, low-level public-safe files for each operational surface:

File	Contents
docs/surfaces/tasks.md	Scheduled tasks/cron jobs, modes, scripts, schedules, delivery class, workdir class.
docs/surfaces/skills.md	Skill categories, counts, public-safe examples, operational semantics.
docs/surfaces/hooks.md	Shell hook allowlist state, webhook state, plugin hook manifests.
docs/surfaces/plugins.md	Hermes plugin registry rows and status.
docs/surfaces/mcp-and-toolsets.md	MCP servers and toolset count estimates.
docs/surfaces/models-and-gateway.md	Model routing, gateway status, channel/platform surface.
docs/surfaces/orchestration.md	Kanban profiles, rules layer, self-improvement automation, and operating contract.

Model Routing

Role	Provider	Model	Notes
Primary	openai-codex	gpt-5.6-terra	Default for Telegram/API/CLI gateway sessions
Fallback			Used when primary fails
Optional provider	lmstudio	qwenvl3bunc	http://127.0.0.1:1234/v1
Optional provider	nvidia	meta/llama-3.3-70b-instruct	https://integrate.api.nvidia.com/v1
Optional provider	freekimi	cfbt-kimi	http://127.0.0.1:3271/v1
Optional provider	forge_freekimi	cfbt-kimi	http://127.0.0.1:8081/v1
Optional provider	forge_lmstudio	qwenvl3bunc	http://127.0.0.1:8082/v1
Optional provider	chatgpt_web	chatgpt-5.6-sol-high-web	https://codex.guber.dev/v1
Optional provider	codex_web_gpt	chatgpt-web/light	http://100.82.137.49:17842/v1

Local model trial status

Item	Status
LM Studio endpoint	not reachable at http://127.0.0.1:1234/v1
Reported model IDs	none
Chat smoke test	blocked_or_unavailable: <urlopen error [Errno 111] Connection refused>
Safety decision	Main Hermes remains openai-codex/gpt-5.5 ; local provider is optional until a model can load reliably.

MCP and External Tooling

MCP server	Transport	Public-safe purpose
gbrain	Local HTTP MCP on 127.0.0.1:3131/mcp	Personal knowledge graph, memory, code graph, facts, schema tools.
notebooklm	npx notebooklm-mcp@latest	Grounded research over registered Google NotebookLM notebooks.
codegraph	Local Node command	Code intelligence over selected repositories.

Scheduled Automation

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Agentic Operating Model

Item	Value	Meaning
Interactive orchestrator	default	Main Telegram/API/CLI profile; decides whether to answer directly, delegate, create Kanban work, or schedule cron.
Research lane	researcher	Evidence-backed OSS/docs/web due diligence and adoption recommendations.
Execution lane	worker	File/terminal/code/config implementation and verification for durable work items.
Review lane	reviewer	Independent review, regression checks, security/config sanity, and final validation.
Default assignee	worker	Fallback when a task/decomposer does not choose a specialist.
Failure limit	5	Raised above the previous aggressive default to avoid noisy retry failures.
Per-profile concurrency	1	Keeps one profile from saturating model/API/browser capacity.
Post-session automation	enabled	Generates review artifacts only; does not auto-install remote skills or mutate active prompts.

The important runtime distinction is that `researcher`, `worker`, and `reviewer` do **not** need to be continuously running gateway profiles. They are idle until a Kanban task is assigned to them; then the gateway dispatcher starts a worker-run for that profile and the worker completes, blocks, or retries the task.

Current Profiles

The live system currently exposes the public-safe profile roster as:

Profile	Model	Gateway	Alias	Distribution
◆default	gpt-5.6-terra	running	–	–
claude	–	stopped	–	–
coding	gpt-5.6-sol	stopped	coding	–
ghidra-restricted	gpt-5.6-sol	stopped	–	–
researcher	gpt-5.6-terra	stopped	hermes-researcher	–
reviewer	gpt-5.6-sol	stopped	hermes-reviewer	–
security-restricted	gpt-5.6-sol	stopped	hermes-security	–
worker	gpt-5.6-terra	stopped	hermes-worker	–

Current profile contract:

Profile	Role	Runtime behavior
default	Interactive orchestrator	Stays running in Telegram/API/CLI gateway; routes work.
researcher	Research/due-diligence worker	Normally stopped; launched by Kanban for evidence gathering.
worker	Implementation/execution worker	Normally stopped; fallback assignee for durable tasks.
reviewer	Independent verification worker	Normally stopped; launched for review/regression/security checks.

Reliability and Safety Boundaries

- Public repo does **not** publish secrets, chat IDs, API keys, OAuth tokens, cookies, private finance details, or client-specific payloads.
- The gateway is supervised by systemd and multiple watchdog cron jobs.
- GBrain runs through a shared local HTTP MCP server to avoid PGLite lock contention.
- Backups and sync jobs run through cron with Google Drive/GitHub targets.
- Windows is treated as a remote worker controlled via SSH, not as the primary Hermes control plane.
- Experimental providers are explicitly optional and must pass direct API + Hermes smoke tests before being used for real agent work.

Operational Notes

- Hermes version/status summary:

```
Hermes Agent v0.21.0 (2026.8.31) · upstream 2b55ded1 · local 3ca096de (+1 carried commit)
Install directory: ~/.hermes/hermes-agent
Install method: git
Python: 3.11.16
OpenAI SDK: 2.24.0
Update available: 679 commits behind – run 'hermes update'
```

- Fallback chain:

```
Primary:  gpt-5.6-terra  (via openai-codex)

Fallback chain (3 entries):
1. ox-alpha-free  (via opencode-go)
2. gpt-5.6-terra  (via openai-codex)
3. gpt-5.3-codex-spark  (via openai-codex)

Tried in order when the primary fails (rate-limit, 5xx, connection errors).
Docs: https://hermes-agent.nousresearch.com/docs/user-guide/features/fallback-providers
```

- MCP list:

MCP Servers:

Name	Transport	Tools	Status
keep-mcp	~/.hermes/scripts/...	16 selected	enabled
codegraph	~/.nvm/versions/no...	all	enabled
gbrain	http://127.0.0.1:3131/mcp	all	enabled
notebooklm	npx -y notebooklm-mcp@latest	all	enabled
windows-cua	~/.local/bin/windo...	all	enabled
monarch	https://api.monarch.com/mcp	17 selected	enabled
cloudflare-api	https://mcp.cloudflare.co...	all	enabled
pixelrag	~/github/pixelrag-...	all	enabled
vibe_trading	~/.hermes/scripts/...	29 selected	enabled
display	https://api.display.dev/v...	all	enabled
open_notebook	~/.hermes/oss-eval...	all	enabled

Maintenance

This repository is regenerated by `scripts/update_architecture.py`. The updater is designed to be safe for a public repository:

1. collect local Hermes runtime state,
2. redact secrets and private identifiers,
3. group sensitive scheduled jobs by category,
4. regenerate Markdown, SVG, HTML, JSON inventory, and PDF,
5. commit and push only if files changed.

Source Files

- `ARCHITECTURE.md` - canonical public architecture document.
- `docs/hermes-architecture.svg` - standalone diagram.
- `docs/hermes-architecture.html` - browser-friendly diagram page.
- `docs/Hermes-Architecture.pdf` - rendered PDF copy.
- `docs/surfaces/*.md` - low-level public-safe inventories for tasks, skills, hooks, plugins, MCP/toolsets, and gateway/model routing.
- `data/inventory.public.json` - redacted machine-readable snapshot.
- `scripts/update_architecture.py` - public-safe regeneration script.

Rendered from GBrain markdown via Hermes brain-pdf.